Validation cross-block responses: z = 0 mm

Identical nominal axes and unit-norm input/output directions; B1 and B4 share the same null vectors.
Observed low-frequency residuals include spring dynamics; they are not an isolated true rigid-body contribution.

B0: nominal static B1: estimated static B4: same static + relative delays

Nominal logical inputs to nullspace output

Response magnitude (dB)

-70 -80 -90 -100 -110 -120 -130 -140 -150

3 10 25 80 170 310

Nullspace input to nominal logical outputs

Response magnitude (dB)

-60 -70 -80 -90 -100 -110 -120 -130 -140 -150

3 10 25 80 170 310

Nullspace input to nullspace output

Response magnitude (dB)

-80 -100 -120 -140 -160 -180 -200 -220

3 10 25 80 170 310

Frequency (Hz)